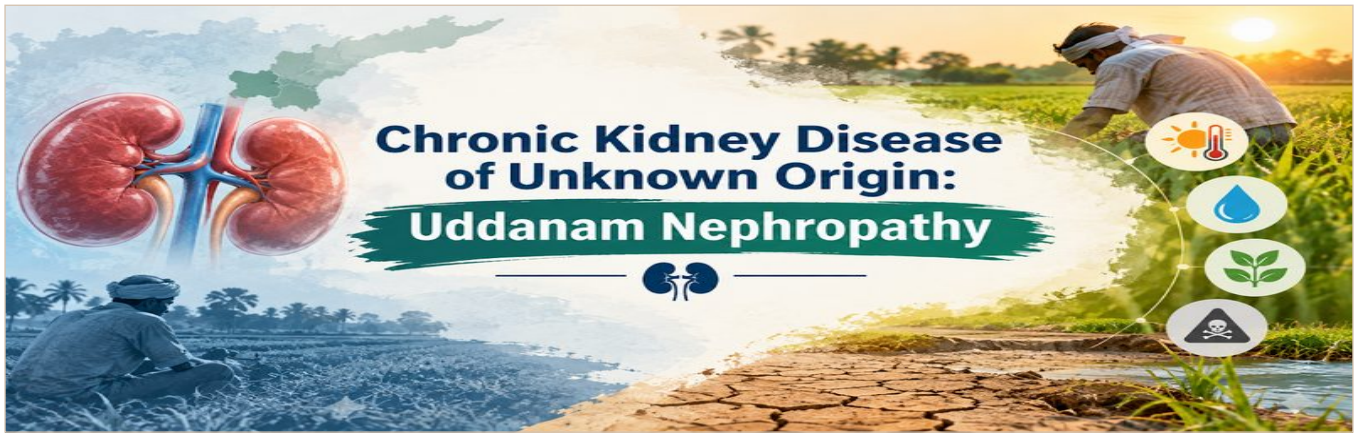


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[GS PAPER II – SOCIAL JUSTICE | • GS PAPER III – ENVIRONMENT & ECOLOGY, SCIENCE & TECHNOLOGY | • ESSAY]
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Chronic Kidney Disease of Unknown Origin: Uddanam Nephropathy



This article examines Uddanam Nephropathy (CKDu), a chronic kidney disease of unknown origin affecting agricultural communities in Andhra Pradesh. It explores the possible links with heat stress, dehydration, environmental exposure and other factors, along with similar kidney diseases reported globally.

Chronic Kidney Disease of Unknown Origin: Uddanam Nephropathy

Syllabus areas:

GS Paper II – Social Justice

GS Paper III – Environment & Ecology, Science & Technology

Essay

1. What is Uddanam Nephropathy?

Uddanam Nephropathy, also referred to as **Chronic Kidney Disease of Unknown Origin (CKDu)**, is a kidney disease reported in the Uddanam region of Andhra Pradesh.

The disease was first reported in the **1990s** and has affected tens of thousands of people in the region, primarily agricultural workers, leading to significant morbidity and mortality.

The exact cause of the disease remains unknown. Patients did not have swollen feet or hands and did not have a history of elevated protein levels in urine.

2. Uddanam and CKDu

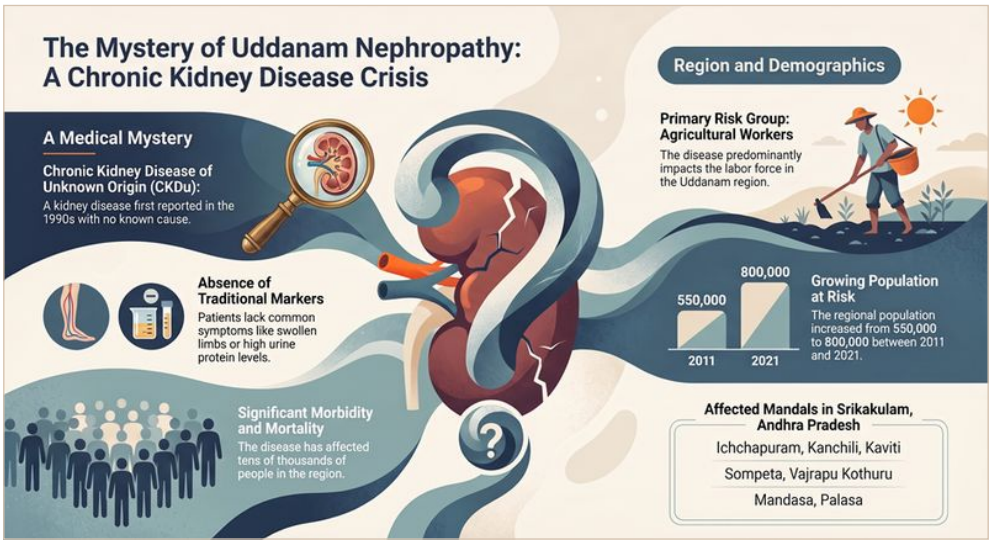
Uddanam is a region in the **Srikakulam district of Andhra Pradesh**, covering the mandals of:

- Ichchapuram
- Kanchili
- Kaviti
- Sompeta
- Vajrapu Kothuru
- Mandasa

· Palasa

The population of the region was over **5,50,000 in the 2011 Census** and **8,00,000 by 2021**.

Uddanam has drawn national and international attention due to the high prevalence of mysterious chronic kidney disease known as **Uddanam Nephropathy/CKDu**.



3. Similar Kidney Diseases Across Different Regions

There are similar forms of kidney disease reported in other agricultural regions of the world.

Balkan Endemic Nephropathy

A similar condition has been reported in the farming villages of Europe's **Balkan Peninsula**.

Scientists found **birthwort**, a plant with a weedy stem growing thick in wheat fields. Compounds called **aristolochic acids** were identified as the cause of severe kidney disease, besides cancer.

This condition is called **Balkan Endemic Nephropathy**.

Mesoamerican Nephropathy

Another such condition is called **Mesoamerican Nephropathy**.

It is:

· Non-diabetic and non-hypertensive chronic kidney disease

· Prevalent in agricultural communities along the Pacific Ocean coastal lowlands

· Reported in Mesoamerica, including:

o Mexico

o Guatemala

o El Salvador

o Nicaragua

o Honduras

o Costa Rica

Sri Lanka and Other Indian Regions

Similar nephropathy has also been reported in **Sri Lanka**.

Coastal agricultural pockets of **Odisha, Goa and Maharashtra** have also reported conditions similar to Uddanam.

In **Tamil Nadu**, 8% of the population has kidney disease, and among them, half has no identifiable cause.

4. Possible Causes

The causes associated with such kidney diseases include:

Extreme Heat and Dehydration

Extreme heat and resulting dehydration are considered possible factors. The focus is on drinking **2 litres of water a day** .

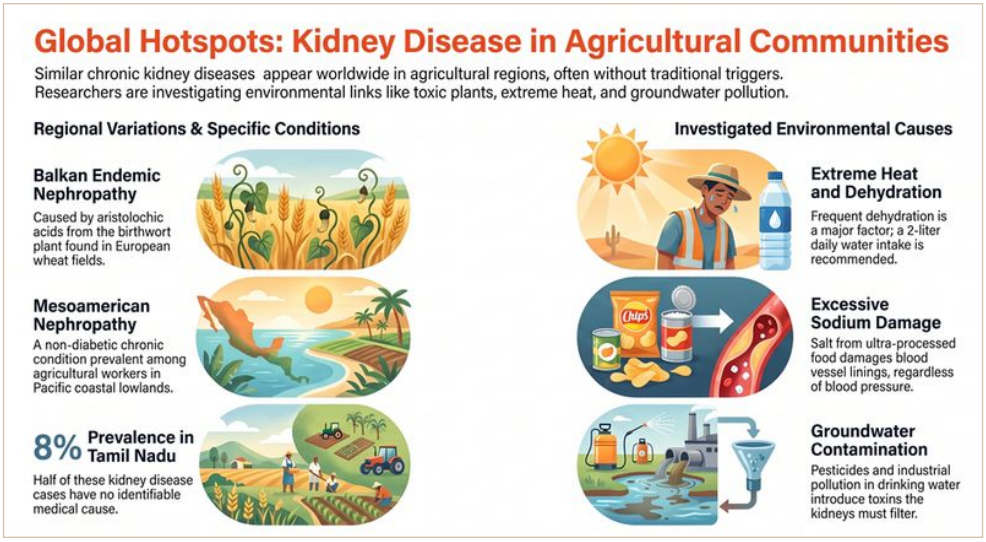
Excessive Salt Intake

Too much salt intake, mostly from ultra-processed food, can lead to excess sodium. Excess sodium can damage the lining of blood vessels, even in people who do not have blood pressure problems.

Groundwater

Groundwater used as drinking water in India has a risk of high salt and metal content, likely caused by **pesticides or industrial pollution**.

The kidneys are particularly vulnerable as they are the primary organs responsible for filtering these toxins from the blood.



5. CKDu and Other Common Kidney Problems

CKDu

In CKDu:

- The cause is unknown.
- Patients generally do not have diabetes or hypertension.
- The disease is particularly associated with young and middle-aged agricultural workers in hot rural areas.
- Suspected links include heat stress, dehydration, agrochemicals and heavy metals.
- Prevention remains unclear, with focus on hydration, reducing agrochemical exposure and avoiding heat stress.
- It is mainly associated with farming regions.

Other Common Kidney Problems

Other common kidney problems usually have a known cause.

CKD – Chronic Kidney Disease can be associated with:

- Diabetes
- Hypertension
- Glomerulonephritis
- Polycystic kidney disease

AKI – Acute Kidney Injury can be associated with:

- Infections

- Toxins
- Dehydration
- Obstruction
- Sepsis

CKD is more common among the elderly, diabetics and hypertensives, while kidney disease can affect anyone and occurs globally.

Measures mentioned include controlling diabetes and hypertension, treating infections and avoiding nephrotoxic drugs.

6. Nephrotoxic Drugs

Some drugs and substances can be harmful to the kidneys, including:

1. Antibiotics

2. Antivirals

3. Pain killers – reduce blood flow to the kidneys by blocking prostaglandins

4. Chemotherapy drugs

5. Immunosuppressants

6. Radiologic contrast agents – used in CT scans and angiography

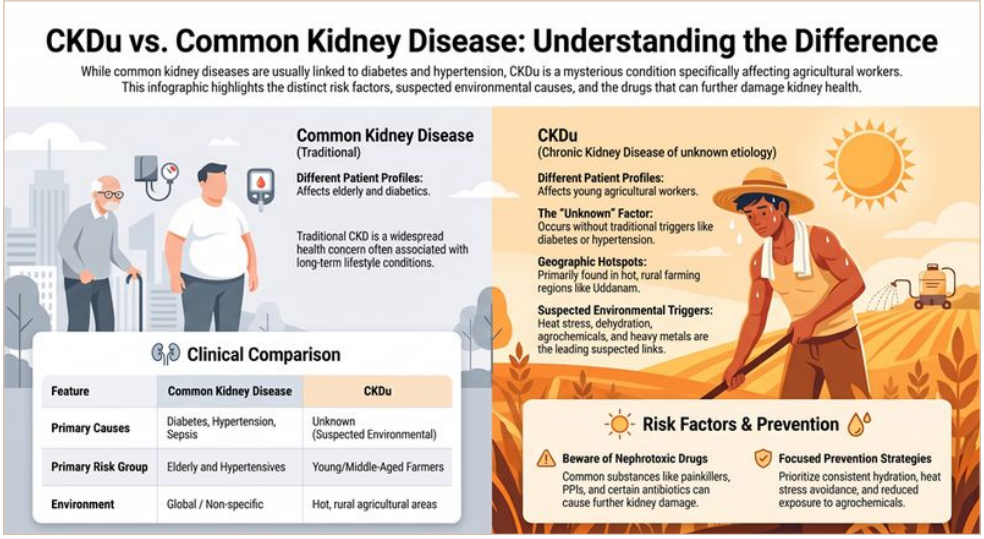
7. Lithium – used in bipolar disorder

8. Proton pump inhibitors – long-term use of Omeprazole, Pantoprazole and Rabeprazole

9. Certain herbal medicines containing aristolochic acid

Conclusion

Uddanam Nephropathy/CKDu represents a form of chronic kidney disease in which the usual causes such as diabetes and hypertension do not explain the condition. Its occurrence among agricultural communities, along with similar kidney diseases reported in other farming regions, has drawn national and international attention. The exact cause remains unclear, with heat stress, dehydration, agrochemicals, heavy metals and other environmental factors among the suspected links.



MCQ's

Q1. A village reports increasing kidney disease among agricultural workers but has relatively low prevalence of conventional metabolic risk factors. Which investigation would be most relevant initially for understanding the cluster?

A. Only screening for diabetes

B. Only measuring blood pressure

C. Assessing occupational heat exposure, hydration, agrochemical use and environmental contaminants along with clinical data

D. Assuming a hereditary disorder because conventional causes are absent

Answer: C

Q2. A disease cluster occurring predominantly among agricultural workers in hot regions necessarily proves that heat exposure is its cause.

A. True B. False

Answer: B

Q3. Consider the following statements regarding nephrotoxic exposures:

- Some analgesics can affect renal blood flow through inhibition of prostaglandin-mediated mechanisms.
- Certain chemotherapeutic agents can have nephrotoxic effects.
- Radiological contrast agents are completely irrelevant to renal function.
- Aristolochic acid exposure has been associated with serious renal disease.

Which of the statements given above are correct?

A. 1 and 2 only B. 1, 2 and 4 only C. 2, 3 and 4 only D. 1, 2, 3 and 4

Answer: B

Q4. Which of the following combinations best represents a One Health/environmental-health perspective on CKDu?

A. Kidney disease !' hospital treatment !' pharmaceutical intervention

B. Agricultural work !' heat exposure !' water stress !' environmental contaminants !' renal risk

C. Diabetes !' obesity !' hypertension !' CKD

D. Infection !' antibiotics !' immunity !' kidney recovery

Answer: B

Q5.

Assertion (A): Occupational heat exposure can be relevant to the study of unexplained kidney disease among agricultural workers.

Reason (R): Repeated dehydration can impose physiological stress on the kidneys, particularly where workers have prolonged exposure to heat.

A. Both A and R are true, and R is the correct explanation of A B. Both A and R are true, but R is not the correct explanation of A C. A is true, but R is false D. A is false, but R is true

Answer: A